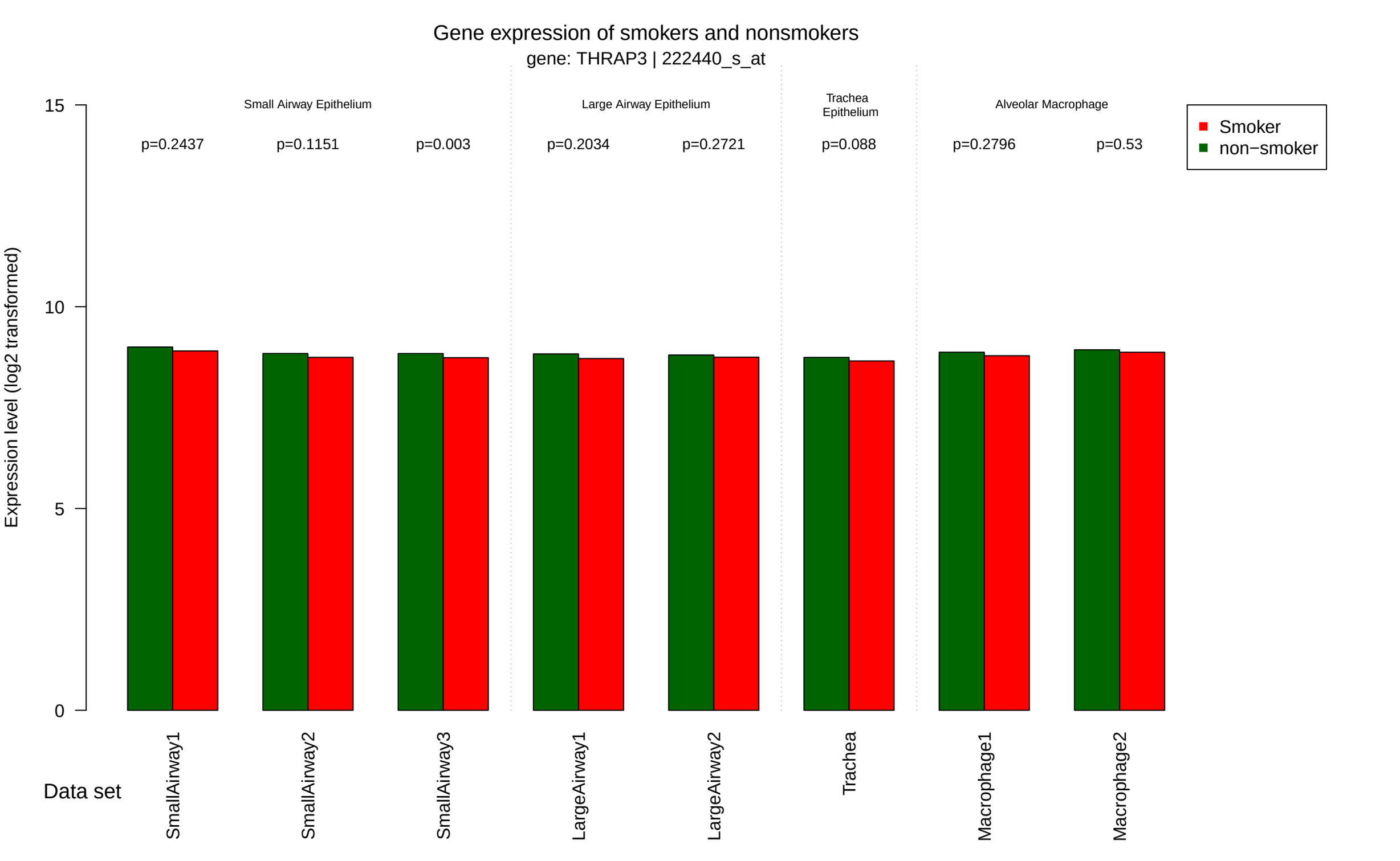
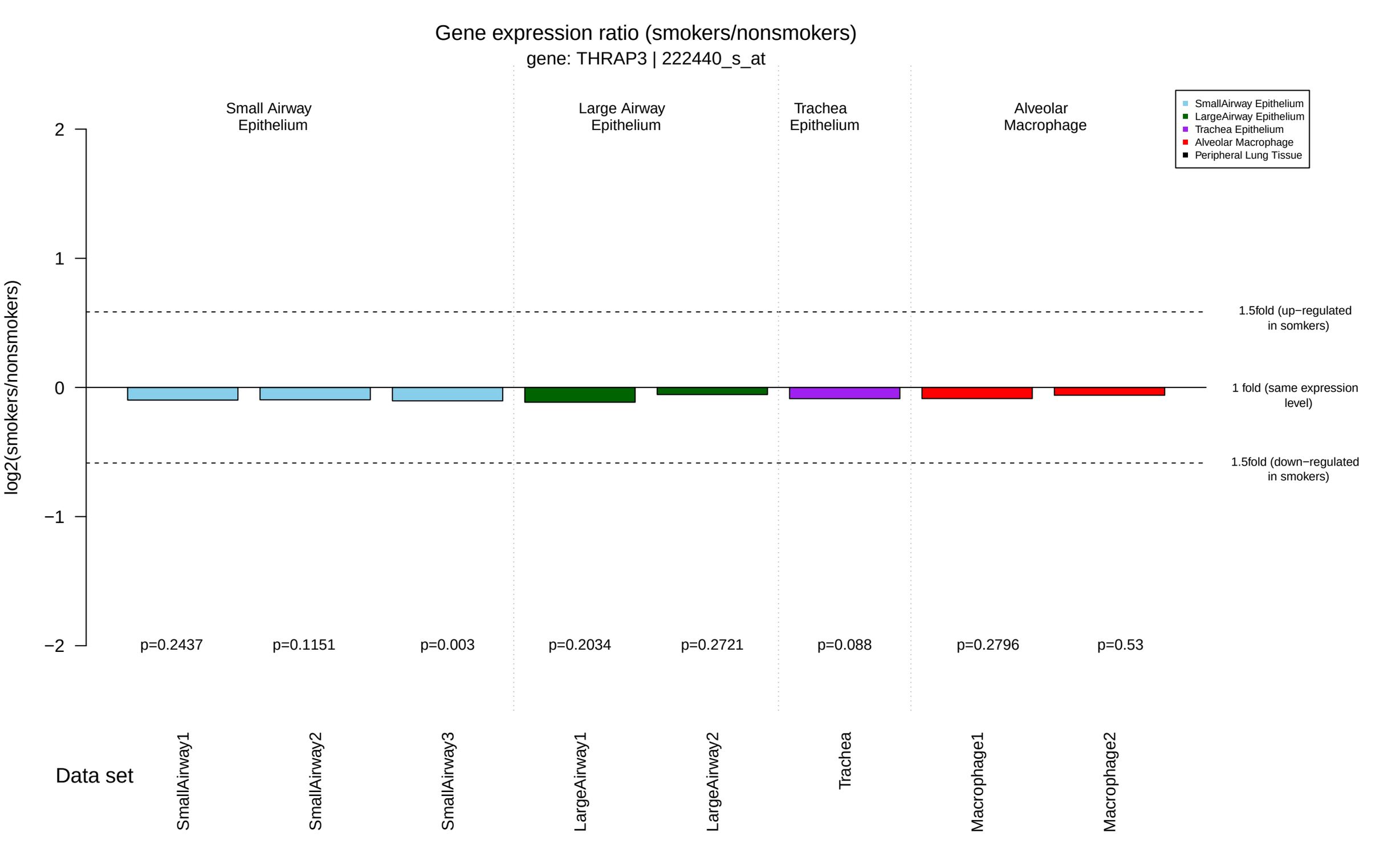
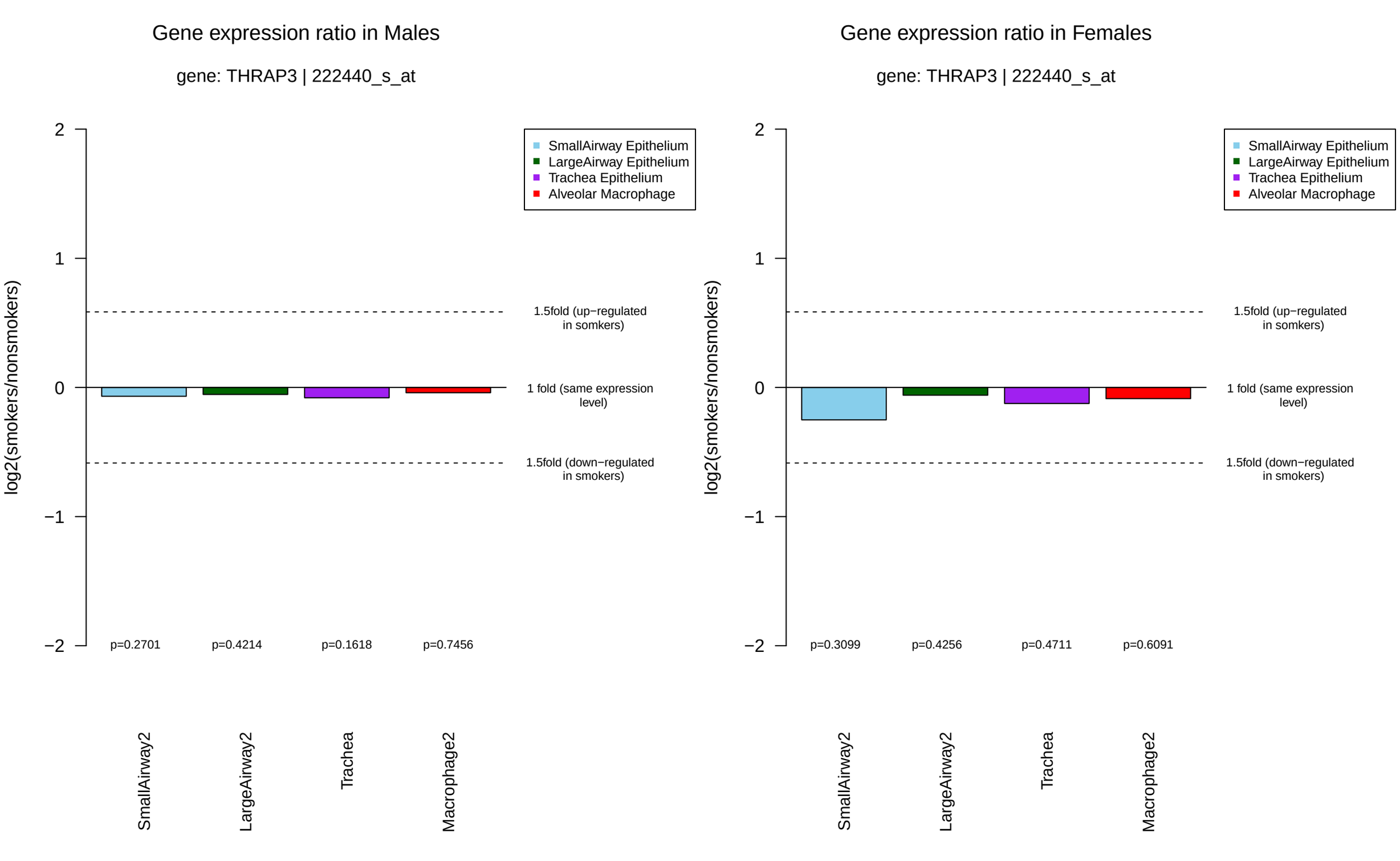
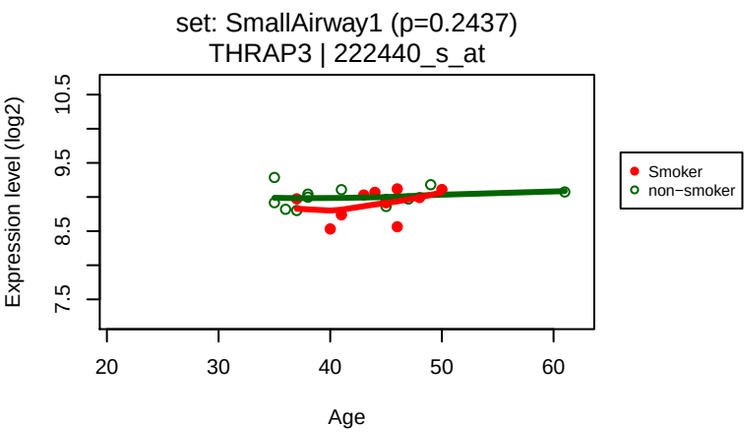
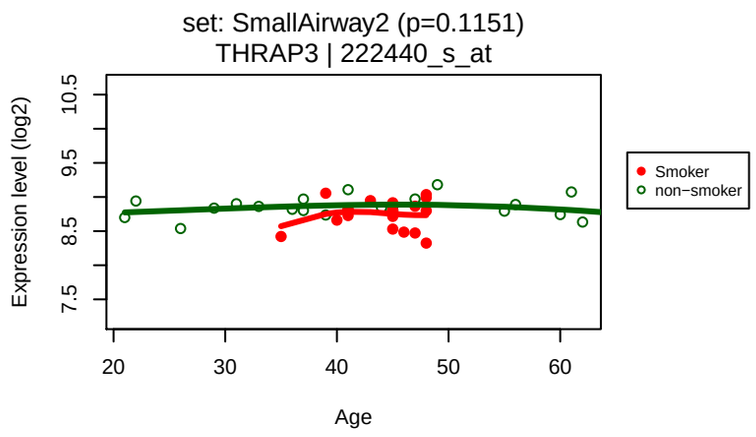
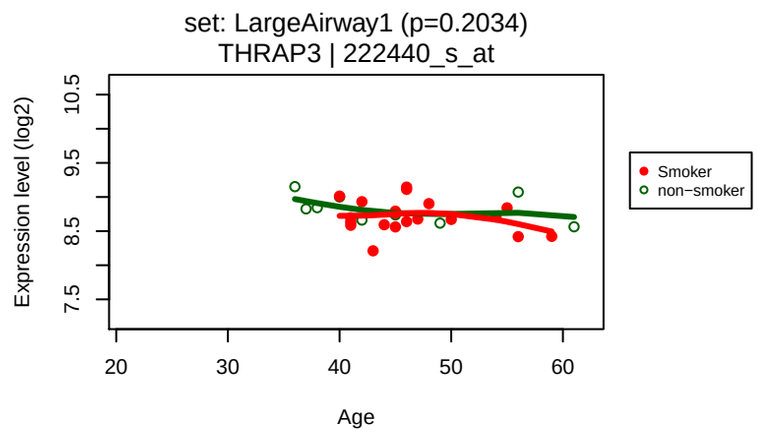

Gene expression of smokers and nonsmokers
gene: THRAP3 | 222440_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2437
p=0.1151
p=0.003
p=0.2034
p=0.2721
p=0.088
p=0.2796
p=0.53
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

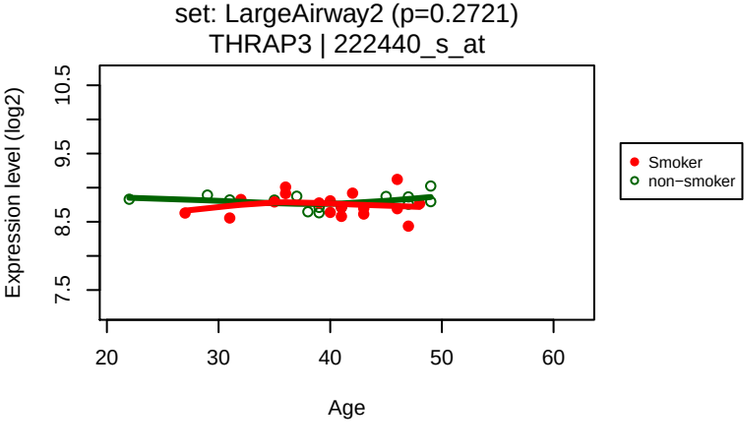

Gene expression ratio (smokers/nonsmokers)
gene: THRAP3 | 222440_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2437
p=0.1151
p=0.003
p=0.2034
p=0.2721
p=0.088
p=0.2796
p=0.53
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2


Gene expression ratio in Males
gene: THRAP3 | 222440_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2701
p=0.4214
p=0.1618
p=0.7456
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: THRAP3 | 222440_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3099
p=0.4256
p=0.4711
p=0.6091
SmallAirway2
LargeAirway2
Trachea
Macrophage2

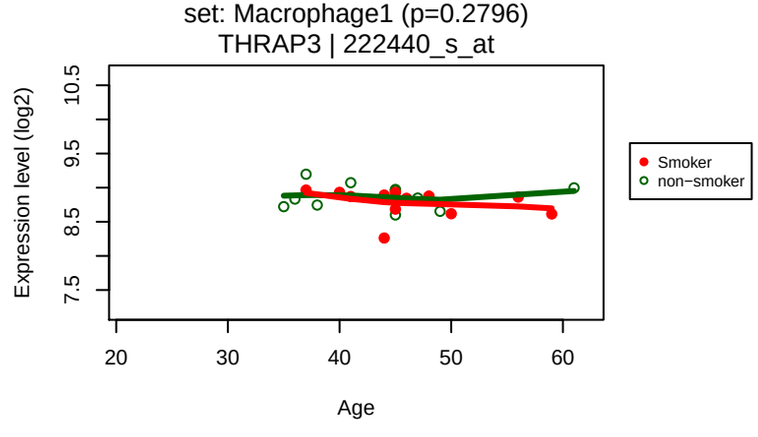

set: SmallAirway1 (p=0.2437)
THRAP3 | 222440_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker

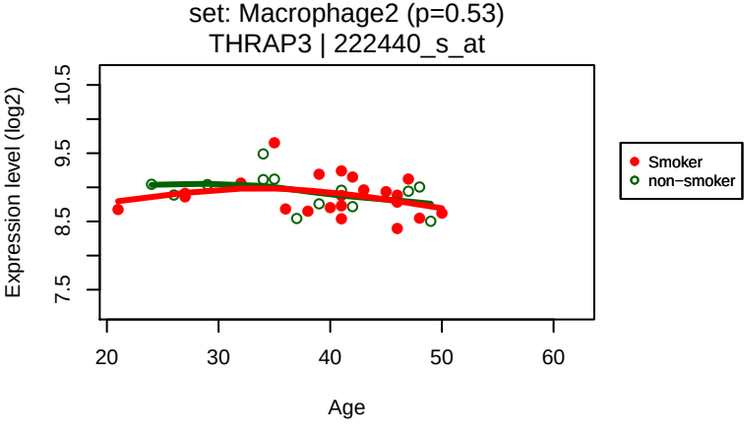

set: SmallAirway2 (p=0.1151)
THRAP3 | 222440_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker

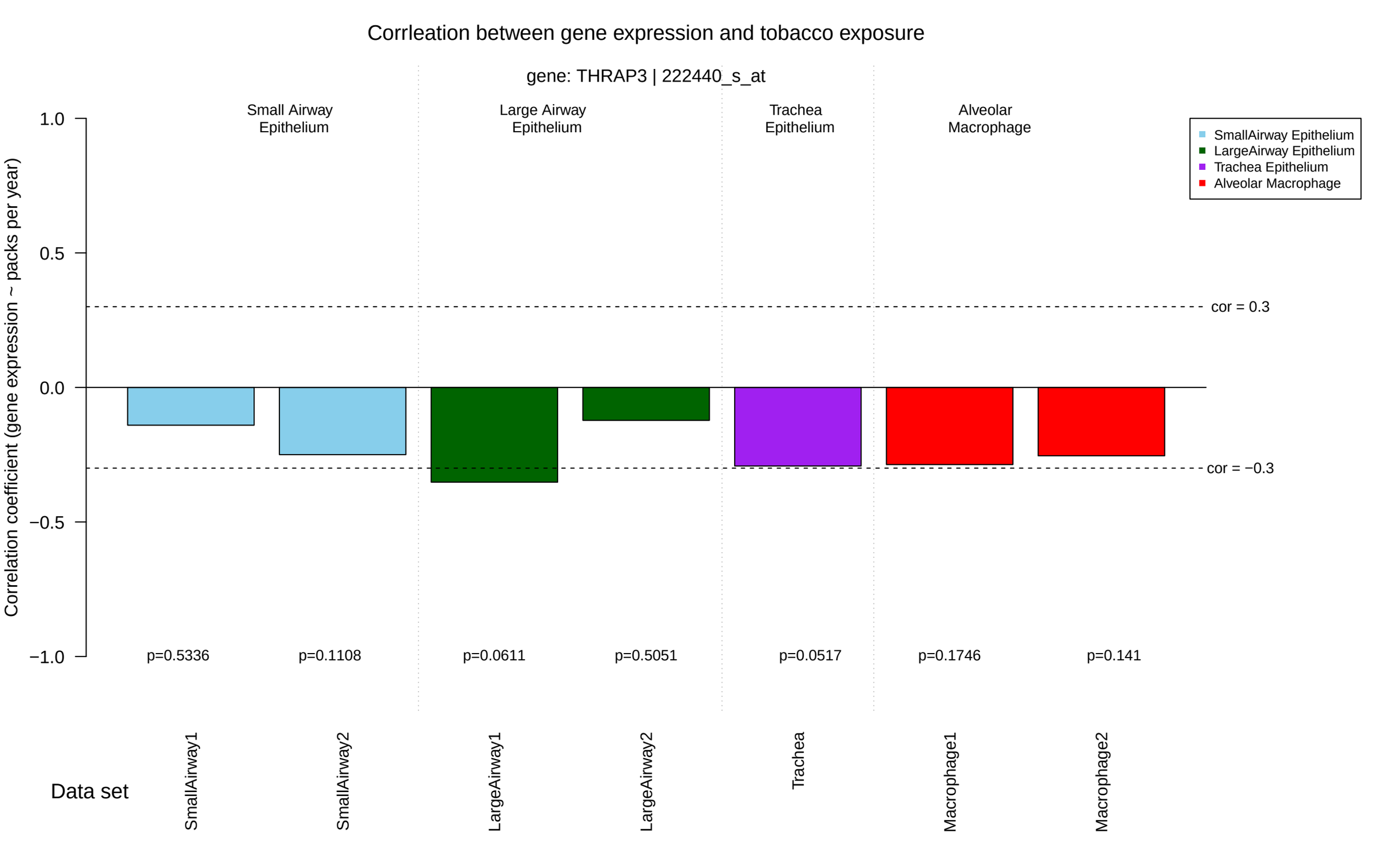

set: LargeAirway1 (p=0.2034)
THRAP3 | 222440_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2721)
THRAP3 | 222440_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker

set: Trachea (p=0.088)
THRAP3 | 222440_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2796)
THRAP3 | 222440_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.53)
THRAP3 | 222440_s_at
Expression level (log2)
7.5
8.5
9.5
10.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: THRAP3 | 222440_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5336
p=0.1108
p=0.0611
p=0.5051
p=0.0517
p=0.1746
p=0.141
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2